



NOMAD

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BASEMAP

Leaflet owns the camera –
42 kB, imperative, plugin-rich.
CARTO dark raster tiles, split
noLabels + **only_labels**, so text
sits **above** the vote heat.

FLATIRON
DISTRICT

WAYPOINTS

One **ordered list of coordinates**

is the whole interface.

The URL is the serialization:

?w=lat,lng;...&vt=<label>

A point **snaps to the walk graph** –

Flatbush R-tree hit-test, $O(\log n)$

over 3.3 M edges. Coordinates survive

a **graph rebuild**; edge ids don't.



ROUTING

The path between waypoints is **derived**, never stored.

OSRM, self-hosted: **foot profile**,
MLD partitioning, sub-ms queries.
annotations=nodes hands back **OSM**
node ids → our edge ids, 1:1.

The votable graph is built from the
same PBF + foot filter as OSRM.



MID WAYPOINTS

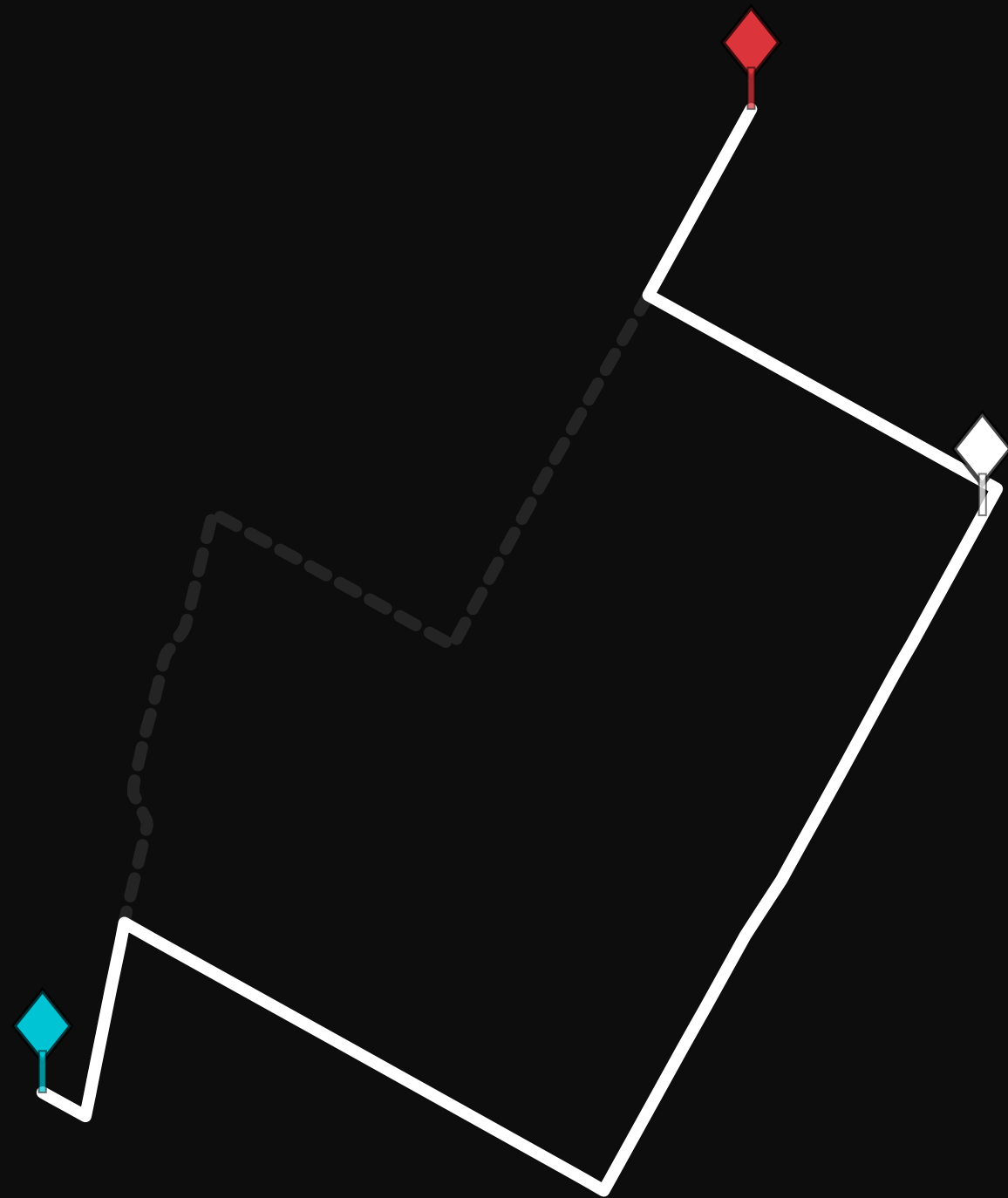
Drag the polyline to pull out a mid;
the route re-derives through it.

Tap vs drag is **time-based** –

TAP_MAX_MS = 300, one feel for the
marker and the path alike.

First is **start**, last is **end**, the
middle **rebalances**. Phase, vote type
and button state are all **derived**.

Tap a kite to delete it – one pure
reducer, no async, 46 unit tests.



BLOCKS

One street is **6–20 OSM edges**:

roadway, both sidewalks, crossing stubs.

Membership is decided **topologically**

first, edge by edge – then each polygon

is generated **from its own members**.

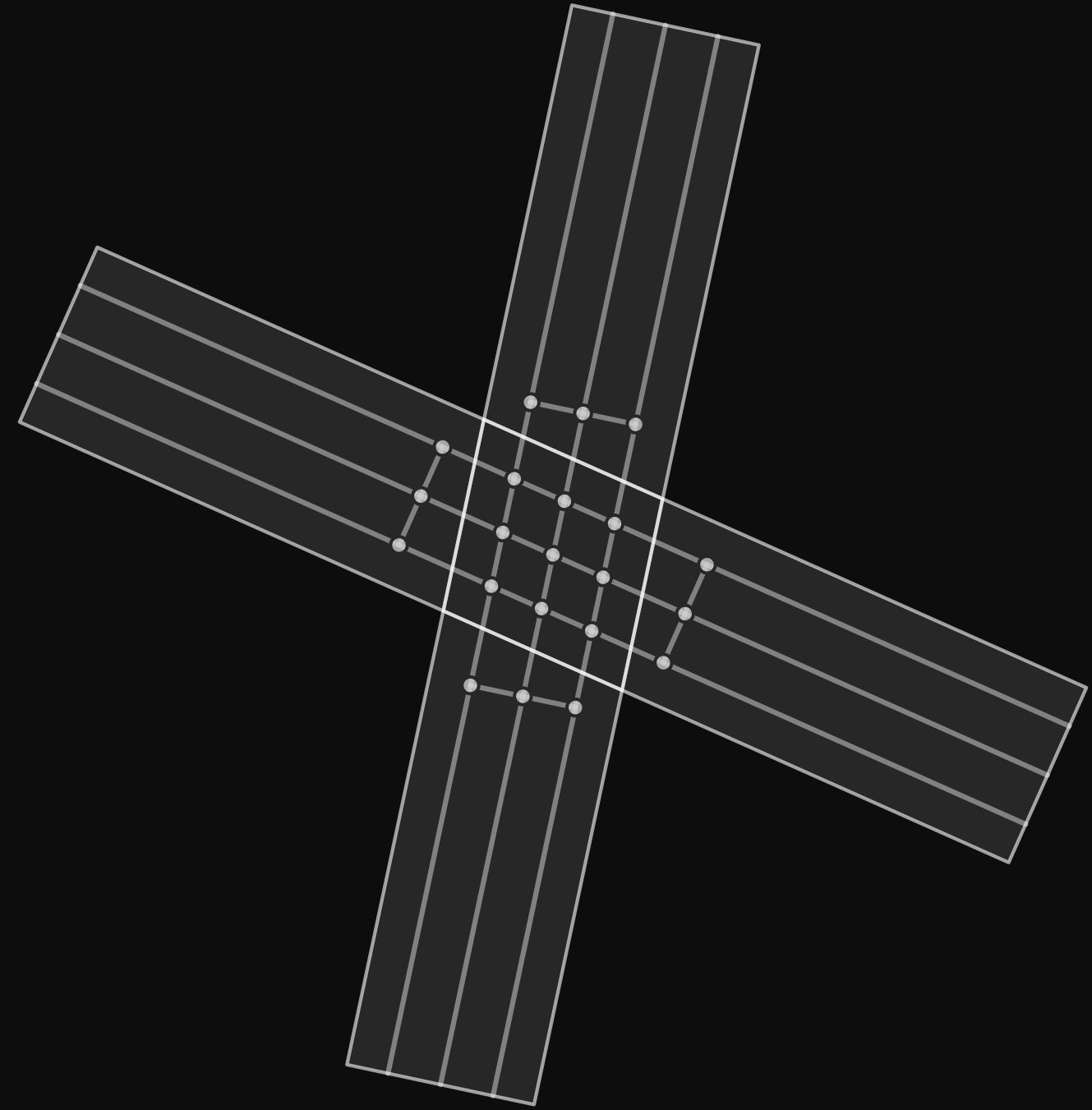
Junction nodes **cluster** into a single

intersection block; a **Voronoi trim**

keeps every block **disjoint**.

Coverage holds **by construction**, and a

device counts **once per block, per type**.



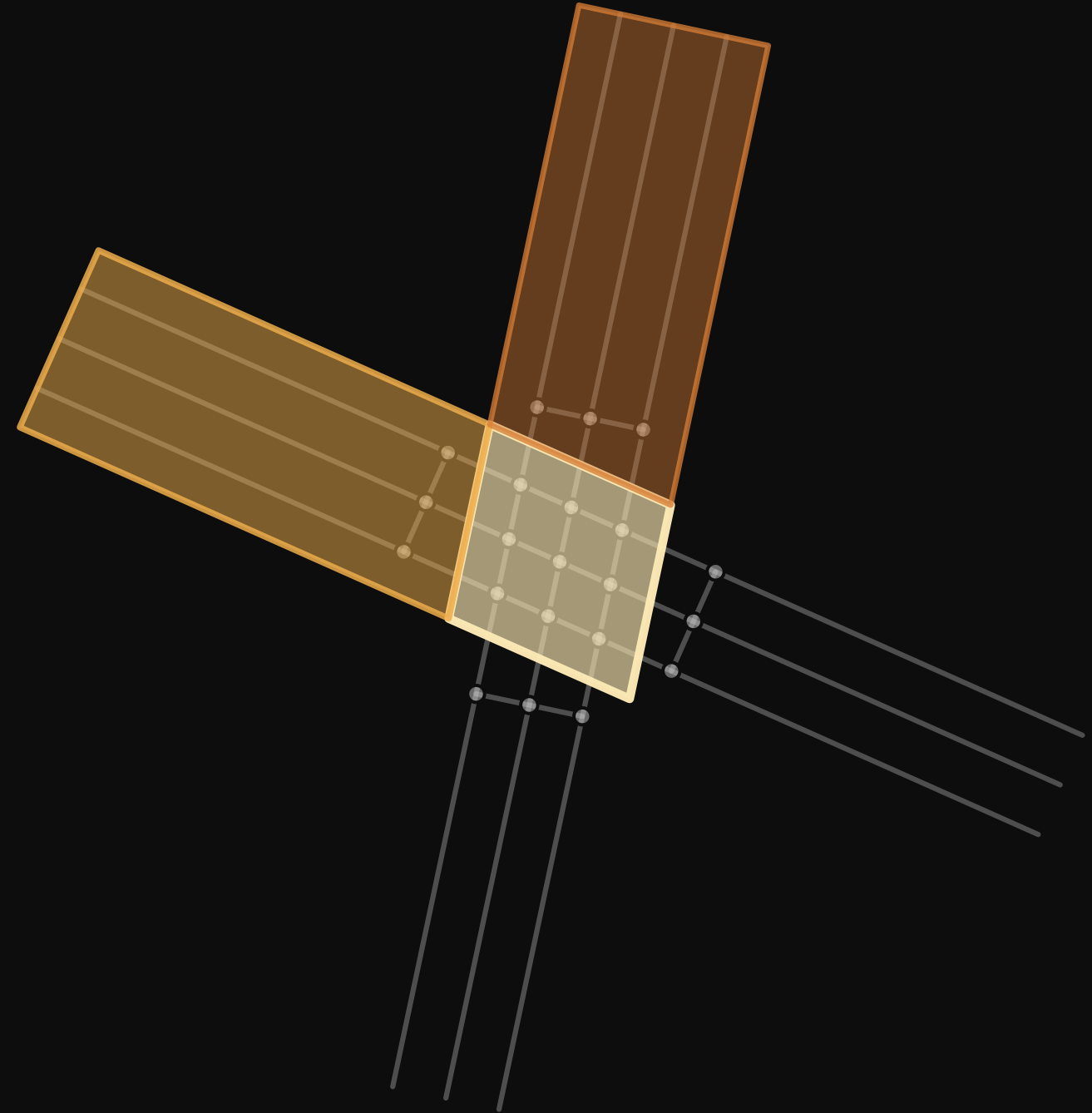
HEAT

A block's heat is its **top proposal's differential** – up minus down.

Log-normalized per arm, with ceilings floored so a quiet map can't saturate.

Warm arm, cold arm, and **zero is invisible** – cancelled signal, no heat.

Applied as **MapLibre feature-state**, diffed per block, never re-styled.



TOP PROPOSAL · POINT

Per block, per **vote type**:

net = **up** – **down**, ranked by differential.

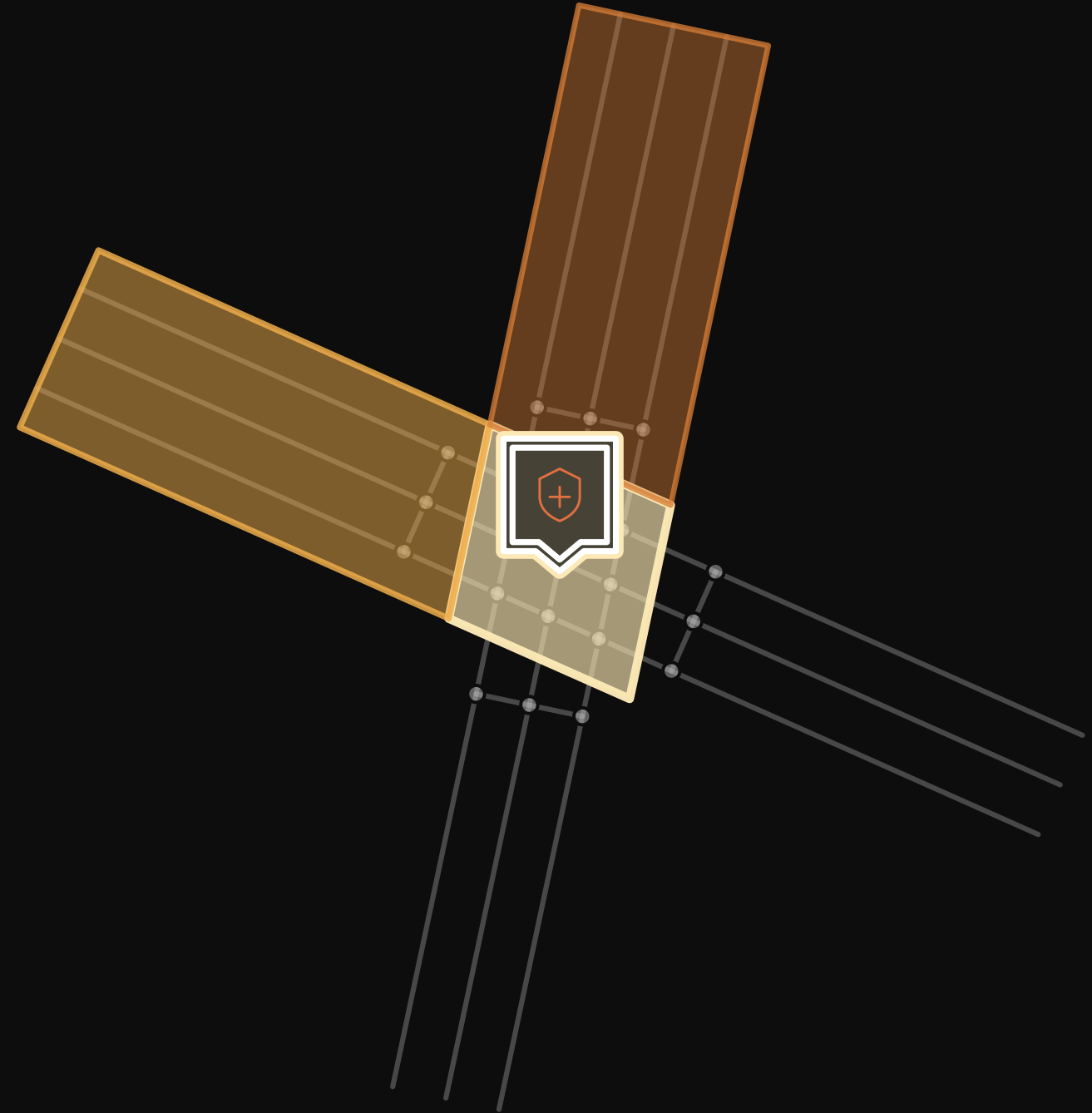
The block's winner becomes **one pin**.

Support floor: **TOP_PROPOSAL_MIN_NET**

= **100**, overridable per map.

6 per type reach the spacing step, then
same-type **non-max suppression at 600 m**
so one avenue can't stack identical pins.

Rank drives the **glow ring** – never the
outline, which belongs to hover.



Edges with **net** ≥ 1 form a per-type subgraph $\rightarrow$ **connected components**.

Peel a corridor from the heaviest edge, taking the strongest arc off either tip.

Budget **2700 m + 660 $\cdot\sqrt{\text{score}}$** , cap 10.5 km

– support buys reach **sublinearly**.

Keep an extension only if routing still **hands the stretch back** ($\geq 85\%$ of blocks); otherwise **pin a ghost waypoint** there.

MAX_GHOST_WAYPOINTS = 3, so a corridor is always ≤ 5 waypoints – shareable, re-routable, **deterministic** on every client.

